GAME THEORY, Spring 2026

Problem Set 4

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1 Strategies and Equilibrium

Consider a two-player game in which player 1 can choose A or B. The game ends if he chooses A while it continues to player 2 if he chooses B. Player 2 can then choose C or D, with the game ending after C and continuing again with player 1 after D. Player 1 can then choose E or F, and the game ends after each of these choices.

a. Model this as an extensive-form game tree. Is it a game of perfect or imperfect information?

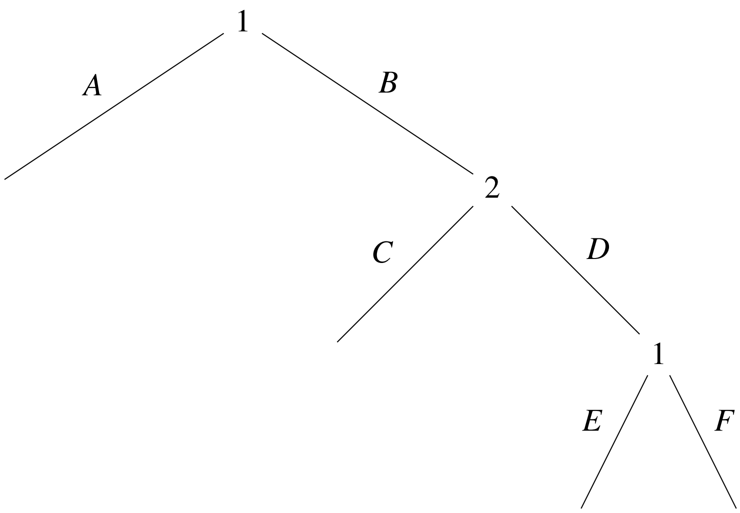
b. How many terminal nodes does the game have? How many information sets?

c. How many pure strategies does each player have?

d. Imagine that the payoffs following choice A by player 1 are $(2, 0)$, those following C by player 2 are $(3, 1)$, those following E by player 1 are $(0, 0)$, and those following F by player 1 are $(1, 2)$. What are the Nash equilibria of this game? Does one strike you as more "appealing" than the other? If so, explain why.

Answer

a. The extensive-form game tree is:



This is a game of perfect information, because at every decision node the player moving knows the full history of previous actions.

b. There are 4 terminal nodes:

$$A, \quad BC, \quad BDE, \quad BDF.$$

There are three decision nodes in total, and since this is a perfect-information game, each decision node is a singleton information set. Therefore the number of information sets is 3.

c. Player 1 has two information sets: at the initial node he chooses between A and B, and at the later node he chooses between E and F. Thus player 1 has $2 \times 2 = 4$ pure strategies:

$$AE, \, AF, \, BE, \, BF.$$

Player 2 has one information set and chooses between C and D, so player 2 has

$$\boxed{2}$$

pure strategies:

$$C, \, D.$$

d. The normal form is

	C	D
AE	$(2, 0)$	$(2, 0)$
AF	$(2, 0)$	$(2, 0)$
BE	$(3, 1)$	$(0, 0)$
BF	$(3, 1)$	$(1, 2)$

For player 1:

- if player 2 chooses C, the best responses are BE and BF;
- if player 2 chooses D, the best responses are AE and AF.

For player 2:

- against AE, both C and D are best responses;
- against AF, both C and D are best responses;
- against BE, C is a best response;
- against BF, D is a best response.

Therefore the pure-strategy Nash equilibria are

$$\boxed{(BE,C),\ (AE,D),\ (AF,D)}.$$

Among these, the most appealing one is

$$\boxed{(AF,D)},$$

because it is the unique subgame-perfect equilibrium.

To see this, use backward induction. At the last node player 1 chooses between E and F, and since

$$1 > 0,$$

he will choose F. Anticipating this, player 2 compares C and D:

$$1 \text{ from } C \quad \text{vs.} \quad 2 \text{ from } D,$$

so player 2 chooses D. Anticipating that, player 1 compares

$$2 \text{ from } A \quad \text{vs.} \quad 1 \text{ from } B,$$

so player 1 chooses A. Hence the backward-induction outcome is (AF,D).

2 Tic-Tac-Toe

The extensive-form representation of a game can be cumbersome even for very simple games. Consider the game of tic-tac-toe, in which two players mark "X" or "O" in a 3×3 matrix. Player 1 moves first, then player 2, and so on. If a player gets three of his kind in a row, column, or one of the diagonals then he wins, and otherwise it is a tie. For this question assume that even after a winner is declared, the players *must completely fill* the matrix before the game ends.

a. Is this a game of perfect or imperfect information? Why?

b. How many information sets does player 2 have after player 1's first move?

c. How many information sets does player 1 have after each of player 2's moves for the first time?

d. How many information sets does each player have in total? (Hint: For this and part (e) you may want to use a spreadsheet program like Excel.)

e. How many terminal nodes does the game have?

3 Veto Power

Two players must choose among three alternatives, a, b, and c. Player 1's preferences are given by

$$a \succ_1 b \succ_1 c$$

while player 2's preferences are given by

$$c \succ_2 b \succ_2 a.$$

The rules are that player 1 moves first and can veto one of the three alternatives. Then player 2 chooses one of the remaining two alternatives.

a. Model this as an extensive-form game tree (choose payoffs that represent the preferences).

b. How many pure strategies does each player have?

c. Find all the Nash equilibria of this game.

4 Brothers

Consider the following game that proceeds in two stages: In the first stage one brother (player 2) has two \$10 bills and can choose one of two options: he can give his younger brother (player 1) \$20 or give him one of the \$10 bills (giving nothing is inconceivable given the way they were raised). This money will then be used to buy snacks at the show they will see, and each \$1 of snacks purchased yields one unit of payoff for a player who uses it. In the second stage the show they will see is determined by the following Battle of the Sexes game:

		Player 2	
		O	F
Player 1	O	$16, 12$	$0, 0$
	F	$0, 0$	$12, 16$

a. Present the *entire game* in extensive form (a game tree).

b. Write the (pure-) strategy sets for both players.

c. Present the *entire game* in one matrix.

d. Find the Nash equilibria of the *entire game* (pure and mixed strategies).

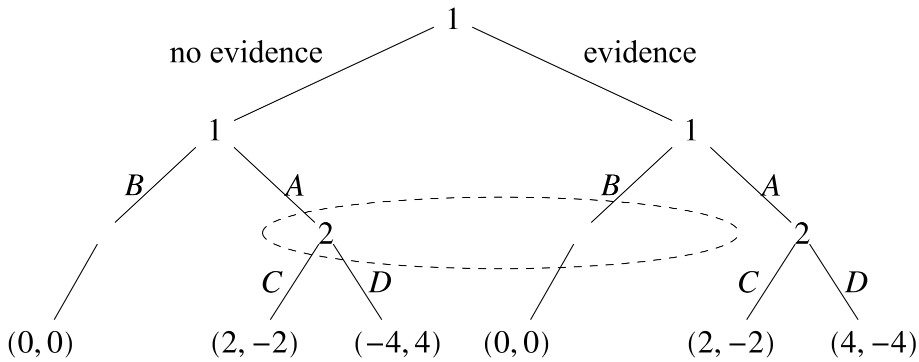
5 The Dean's Dilemma

A student stole the DVD player from the student lounge. The dean of students (player 1) suspects the student (player 2) and begins collecting evidence. However, evidence collection is a random process, and concrete evidence will be available to the dean only with probability $\frac{1}{2}$. The student knows that the evidence-gathering process is under way but does not know whether the dean has collected evidence or not. The game proceeds as follows: The dean realizes whether or not he has evidence and can then choose his action, whether to accuse the student (A) or bounce the case (B) and forget it. Once accused, the student has two options: he can either confess (C) or deny (D). Payoffs are realized as follows: If the dean bounces the case then both players get 0. If the dean accuses the student and the student confesses, the dean gains 2 and the student loses 2. If the dean accuses the student and the student denies, then payoffs depend on the evidence: If the dean has no evidence then he loses face, which gives him a loss of 4, while the student gains glory, which gives him a payoff of 4. If, however, the dean has evidence then he is triumphant and gains 4, while the student is put on probation and loses 4.

a. Draw the game tree that represents the extensive form of this game.

b. Write down the matrix that represents the normal form of the extensive form you drew in (a).

c. Solve for the Nash equilibria of the game.



6 Centipede Game

Consider the following normal form representation of the centipede game we covered in class. Find all Nash equilibria.

	nn	nc	cn	cc
NN	$1, 1$	$1, 1$	$1, 1$	$1, 1$
NC	$1, 1$	$1, 1$	$1, 1$	$1, 1$
CN	$0, 3$	$0, 3$	$2, 2$	$2, 2$
CC	$0, 3$	$0, 3$	$1, 4$	$3, 3$